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Muscle-to-Bone Ratio in NCAA Division I Collegiate Football Players by Position

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[1]School of Kinesiology, University of Minnesota, Minneapolis, Minnesota; [2]Red Bull, Performance and Data Analytics, Athlete Performance Center, Santa Monica, California; [3]Department of Dietetics and Nutrition, University of Kansas Medical Center, Kansas City, Kansas; and [4]Department of Kinesiology and Health Education, University of Texas at Austin, Austin, Texas

Abstract

Dengel, DR, Studee, HR, Juckett, WT, Bosch, TA, Carbuhn, AF, Stanforth, PR, and Evanoff, NG. Muscle-to-bone ratio in NCAA division I collegiate football players by position. *J Strength Cond Res* XX(X): 000–000, 2024—The purpose of this study was to compare the muscle-to-bone ratio (MBR) in National Collegiate Athletic Association Division I football players (collegiate football players [CFP]) to healthy, age-matched controls. In addition, we examined MBR in CFP by position. A total of 553 CFP and 261 controls had their total and regional lean mass (LM), fat mass (FM), and bone mineral content (BMC) determined by dual x-ray absorptiometry (DXA). College football players were categorized by positions defined as offensive linemen (OL), defensive linemen (DL), tight end, linebacker (LB), running back (RB), punter or kicker, quarterback (QB), defensive back (DB), and wide receiver (WR). There were significant differences between CFP and controls for total LM (80.1 ± 10.0 vs. 56.9 ± 7.8 kg), FM (22.2 ± 12.5 vs. 15.2 ± 7.1 kg), and BMC (4.3 ± 0.5 vs. 3.1 ± 0.5 kg). Although there were significant differences in body composition between CFP and controls, there was no significant differences in total MBR between CFP and controls (18.6 ± 1.4 vs. 18.8 ± 1.7). Regionally, CFP had significantly lower trunk MBR than controls (26.7 ± 2.7 vs. 28.7 ± 4.2), but no difference was seen in leg or arm MBR. Positional differences in CFP were noted as total MBR being significantly higher in DL (19.0 ± 1.4) than in DB (18.1 ± 1.3), WR (18.1 ± 1.3), and LB (18.2 ± 1.3). OL had a significantly higher total MBR (19.2 ± 1.3) than DB (18.1 ± 1.3), LB (18.2 ± 1.3), QB (18.1 ± 1.0), and WR (18.1 ± 1.3). In addition, RB had significantly higher total MBR (18.8 ± 1.3) than DB (18.1 ± 1.3) and WR (18.1 ± 1.3). This study may provide athletes and training staff with normative values when evaluating total and regional MBR with DXA.

Key Words: dual x-ray absorptiometry, body composition, athletics, percent fat

Introduction

Previously, in professional (5,9) and collegiate (6,28) American football players, we have demonstrated that specific body composition characteristics may help identify success in the sport. In addition, we (6,9,23) and others (18,26) have demonstrated that football player's body composition varies by position. These positional differences in body composition may be a result of the different demands of each position, ultimately contributing to different total body and regional (arms, legs, and trunk) tissue distributions (6,11).

Bones and muscles are biomechanically linked (21). Thus, evaluating bone and muscle as a functional unit is valuable, particularly in situations where 1 or both tissues may be adversely or beneficially affected. In humans, the muscle-to-bone ratio (MBR) is a metric that examines the ratio between muscle and bone masses and may provide a way to examine the link between muscles and bone. This relationship has been previously examined in a number of sports (3,7,8,13,15). Although dual x-ray absorptiometry (DXA) is more accurate than anthropometrics to determine body composition, very few studies (10,12,16) have used DXA to determine the MBR. Recently, we used DXA to compare regional and total MBR in professional football players in the National Football League (NFL) to a healthy age-matched control group (10). The total and trunk MBR was lower in the NFL players compared with the healthy age-matched control group (10). When NFL players were categorized by their offensive or defensive position for comparison, no differences in total MBR were observed. However, leg MBR varied greatly among NFL players in respect to positions. Positions that mirrored each other showed similar leg MBR. Although we have previously reported total and regional MBR in NFL players (10), these values may not accurately reflect MBR values in collegiate football players (CFP). During a football player's collegiate career, they are often expected to add additional weight. In addition, during these years, men may still be growing and increasing bone mass (19). Both of these factors may result in differing MBR values between CFP and NFL players. Information regarding CFP MBR may provide coaches, performance staff, and players with a way to monitor changes in muscle and bone because of growth, as well as strength programs and nutrition plans in athletes. Thus, the primary purpose of this study was to examine DXA-derived total and regional measures of MBR in National Collegiate Athletic Association (NCAA) Division I CFP and compare them with a healthy age-matched control group. We hypothesized that total MBR would not differ between CFP and controls, but there would be regional MBR differences. The secondary aim was to evaluate total and regional MBR between the positions.

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We also hypothesized that both total and regional MBR would differ between positions in CFP.

Methods

Experimental Approach to the Problem

This study used a retrospective analysis from previously collected data on NCAA Division I CFP (6). Data from CFP were compared with an age-matched, sex-matched, and body mass index (BMI)–matched healthy control group. The CFP and healthy control group both consisted of men between the age of 18 and 24 years from the University of Minnesota, University of Kansas, University of Texas at Austin, and Texas Christian University. Because of the size limitations of the DXA scanner table, not all individuals were able to fit into the DXA scanning field area. Therefore, scan data were only used for those individuals who fit 100% in the DXA scanning field area.

Subjects

A total of 553 NCAA Division I CFP were compared with 261 healthy age-matched, sex-matched, and BMI-matched college controls, who were registered for a physical fitness class meeting 2 times per week. College football players were categorized by positions for analysis, including offensive linemen (OL), defensive linemen (DL), tight ends (TE), linebackers (LB), running backs (RB), punters and kickers (PK), quarterbacks (QB), defensive backs (DB), and wide receivers (WR).

This study was approved by the University of Minnesota Institutional Review Board; the data transfer from other universities was approved by each individual university. All subjects were informed of the benefits and risks of the investigation before signing an institutionally approved informed consent document to participate in the study. For individuals younger than 18 years, parental or guardian consent was also obtained.

Procedures

For CFP, only DXA scans conducted during the preseason were included (June 1–August 1) and were performed on off-workout days or at least 2 hours after a practice session. For the age-matched, sex-matched, and BMI-matched control group DXA scans were completed during the academic calendar (September 1- June 1). Before testing, subjects were directed to remove any jewelry, metal items, and excess clothing. All subjects were instructed to maintain their hydration status before testing. Height and body mass were measured by a standard stadiometer and electronic scale for all subjects, respectively. Body mass index was calculated by dividing weight in kilograms by height in meters squared. Body composition was determined on each subject with a whole-body DXA (iDXA/Lunar Prodigy; General Electric Medical Systems, Madison, WI) scan using standard DXA procedures. All subjects were positioned supine and centered on the scanning table, making sure as much of their body was in the scanning field as possible. All scans were analyzed using the 16.2 version of enCore software by the same technician to ensure that points of interest were correctly positioned. Regional measures included arms, legs, and trunk. Total and regional MBR were then calculated by dividing total or regional lean mass (LM) by total or the corresponding regional bone mineral content (BMC). The reliability and validity of DXA on football players has been demonstrated previously (4).

Table 1

Mean (±*SD*) total body composition measures of collegiate football players (CFP) and age-matched controls.*

	Control (n = 261)	Collegiate football players (n = 553)	p
Total percent fat (%)	19.8 ± 6.7	19.6 ± 7.5	0.672
Total fat mass (kg)	15.2 ± 7.1	22.2 ± 12.5	<0.001
Total lean mass(kg)	56.9 ± 7.8	80.1 ± 10.0	<0.001
Total BMC (kg)	3.05 ± 0.48	4.32 ± 0.50	<0.001
Total BMD ($g \cdot cm^{-2}$)	1.27 ± 0.13	1.60 ± 0.12	<0.001

*BMC = bone mineral content; BMD = bone mineral density.

Statistical Analyses

Descriptive statistics were calculated using mean and *SD* of each variable of interest. Two sample t-tests were used to test for differences in each variable of interest between the CFP control groups. Analysis of variance tests were used to assess whether positional mean values were equal to each other. Bonferroni's corrections were used to adjust for multiple comparisons. All data were analyzed using R Studio (R Foundation for Statistical Computing, Version 4.1.3, Vienna, Austria) and GraphPad Prism statistical software (GraphPad Software, San Diego, California). Statistical significance was determined with an alpha of $p \leq 0.05$.

Results

Five hundred fifty-three CFP were matched to a population of healthy men ($n = 261$) by age. Therefore, by design, there were no significant differences in age (20.1 ± 1.5 vs. 20.4 ± 1.4 years, $p = 0.071$) between CFPs and controls. The CFP were significantly ($p < 0.001$) taller (186.1 ± 6.7 cm vs. 175.6 ± 6.8 cm), heavier (106.8 ± 21.3 kg vs. 74.9 ± 11.9 kg), and had a greater BMI (30.6 ± 5.0 vs. 24.2 ± 3.4 $kg \cdot m^{-2}$) compared with the control group. Dual x-ray absorptiometry body composition results of CFP and control subjects are displayed in Table 1.

College football players had significantly ($p < 0.001$) greater fat mass (FM), LM, BMC, and bone mineral density (BMD) than controls. However, the total percent body fat (%BF) was not significantly different between CFP and controls. Similar to total body composition measures, CFP had significantly ($p < 0.001$) greater amounts of arm, leg, and trunk measures of FM, LM, and BMC than controls (Table 2). There was no significant ($p = 0.087$) difference in total MBR between CFP and controls (18.6 ± 1.4 vs. 18.8 ± 1.7) (Figure 1). Similarly, there were no significant

Table 2

Mean (±*SD*) regional body composition measures of collegiate football players and age-matched controls.*

	Control (n = 261)	CFP (n = 553)	p
Arm fat mass (kg)	1.50 ± 0.70	2.30 ± 1.26	<0.001
Arm lean mass (kg)	7.63 ± 1.83	11.38 ± 1.69	<0.001
Arm BMC (kg)	0.44 ± 0.08	0.66 ± 0.09	<0.001
Leg fat mass (kg)	5.40 ± 2.23	7.70 ± 3.90	<0.001
Leg lean mass (kg)	19.70 ± 2.97	28.90 ± 4.47	<0.001
Leg BMC (kg)	1.17 ± 0.20	1.70 ± 0.19	<0.001
Trunk fat mass (kg)	7.53 ± 4.44	11.18 ± 7.52	<0.001
Trunk lean mass (kg)	26.10 ± 3.75	36.14 ± 4.21	<0.001
Trunk BMC (kg)	0.91 ± 0.19	1.37 ± 0.21	<0.001

*BMC = bone mineral content.

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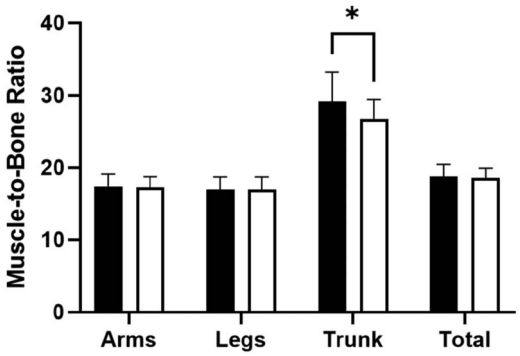

Figure 1. Mean (±*SD*) total and regional muscle-to-bone ratio of controls (solid bars) and collegiate football players (open bars). *Significantly ($p < 0.05$) different.

differences in leg (17.0 ± 1.8 vs. 16.9 ± 1.8, $p = 0.859$) or arm (17.3 ± 1.5 vs. 17.4 ± 1.7, $p = 0.391$) MBR (Figure 1). However, trunk MBR was significantly lower in CFP compared with controls (26.7 ± 2.7 vs. 28.7 ± 4.2, $p < 0.001$) (Figure 1).

There were significant positional differences in the physical characteristics in the CFP group. Offensive linemen were significantly taller than all positions, except TE (Table 3). In addition, OL were also significantly heavier and had a higher BMI than all other positions (Table 3). By contrast, WR weighed significantly less than most positions, excluding PK or DB. Punters and kickers had a significantly lower BMI than most positions, excluding QB, WR, and DB (Table 3).

Positional differences in total body composition included OL having significantly greater FM and %BF than DL, but the 2 were similar for all measures of LM, BMC, and BMD (Table 4). Positions that mirrored each other, including TE and LB as well as DB and WR, were not significantly different from each other in FM, LM, BMC, or BMD. Similarly, PK and QB were not significantly different from each other in FM, LM, BMC, or BMD. Both OL and DL had significantly more LM and BMC than all other positions. Defensive back had significantly less FM than most positions, except PK and WR.

In CFP, significant differences in total MBR were observed between the positions (Figure 2, Panel A). DL and OL had a significantly greater total MBR (19.0 ± 1.4 and 19.2 ± 1.3, respectively) than DB (18.1 ± 1.3, $p < 0.001$), WR (18.1 ± 1.3, $p < 0.001$), and LB (18.2 ± 1.3, $p = 0.001$). In addition, OL had a significantly greater total MBR (19.2 ± 1.3) than QB (18.1 ± 1.0, $p = 0.007$). Finally, RB had a significantly higher total MBR (18.8 ± 1.3) than DB (18.1 ± 1.3, $p = 0.038$) and WR (18.1 ± 1.3, $p = 0.022$) but not PK (18.6 ± 1.1).

We also observed significant differences in regional body composition between the positions (Table 5). Similar to total body composition, OL had a significantly greater amount of arm, leg, and trunk FM than DL but were similar for all regional measures of LM and BMC, except for trunk BMC. OL also had significantly more FM in each region compared with every other position. In addition, similar to total body composition results, position pairings of TE and LB as well as DB and WR mirrored each other for all regional measures of FM, LM, and BMC. Punters and kickers and QB also had similar regional measures of FM, LM, and BMC.

There were significant positional differences in arm, leg, and trunk MBR, which are presented in Figure 2 (Panels B, C, and D). Arm MBR was significantly lower in DB (16.8 ± 1.3) and WR (16.9 ± 1.3) than in DL (18.0 ± 1.7, $p < 0.001$), OL (17.6 ± 1.7, $p < 0.05$), and RB (17.8 ± 1.1, $p = 0.05$). In addition, LB had a significantly lower arm MBR (17.0 ± 1.1) than DL (18.0 ± 1.7, $p = 0.007$). There were a number of differences in leg MBR: OL had significantly higher leg MBR than every other position. Punters and kickers had significantly lower leg MBR (15.6 ± 1.2) than DL (18.0 ± 1.7, $p < 0.001$), OL (18.6 ± 1.7, $p < 0.001$), and RB (17.4 ± 1.4, $p = 0.012$). For trunk MBR, OL (25.6 ± 2.5) was significantly lower than DB (27.5 ± 2.9, $p < 0.001$), PK (28.5 ± 2.5, $p = 0.006$), RB (27.5 ± 2.6, $p = 0.001$), and WR (27.3 ± 2.8, $p < 0.001$). There were no significant differences in regional MBR between positions that mirror each other, except in leg MBR between OL and DL.

Discussion

The primary purpose of this study was to examine DXA-derived total and regional measures of MBR in NCAA Division I CFP and compare these measures to a healthy age-matched control group. To the best of our knowledge, this is the first study to explore total and regional MBRs in the NCAA Division I CFP. In this study, no significant difference was seen in total MBR between CFP and control subjects. This lack of difference may be because of the wide variation of body composition seen in CFP, which is similar to the variation of body composition seen in the control group. This lack of difference may also be because of CFP having significantly higher LM and BMC compared with controls. Because both variables were elevated in CFP, the MBR remained consistent. It is possible that the physiological demands and strength training required in college football may impact both LM and BMC, as BMC is influenced by mechanical forces and muscular adaptations (1,20,24). Animal models support that both muscle and bone functional capacities are affected by physical activity (27).

In this study, we observed no significant differences in total MBR between CFP and controls. These results differ from previous research in NFL players (10), which identified a lower total MBR in NFL players compared with controls. Similar to the current study's CFP, the NFL players had more total LM and BMC compared with their respective control group. However, the proportion of LM to BMC in NFL players was lower compared with their controls (10), whereas CFP in this study had similar proportions to their control group. This may be because of

Table 3

Mean (±*SD*) physical characteristics of collegiate football players by position.*†

Position	OL (n = 125)	DL (n = 67)	TE (n = 38)	LB (n = 62)	RB (n = 48)	PK (n = 13)	QB (n = 25)	DB (n = 85)	WR (n = 90)
Age (y)	20.1 ± 1.4	20.3 ± 1.5	20.1 ± 1.3	20.2 ± 1.5	20.3 ± 1.5	20.3 ± 1.6	19.6 ± 1.6	20.1 ± 1.4	19.8 ± 1.4
Height (cm)	192.6 ± 5.3^{d}	188.2 ± 3.8^{b}	190.1 ± 5.1^{bd}	184.8 ± 3.7^{ce}	179.3 ± 3.8^{a}	187.4 ± 4.1^{bce}	187.1 ± 4.9^{bc}	180.9 ± 4.2^{af}	183.0 ± 6.4^{ef}
Mass (kg)	135.8 ± 9.3^{d}	123.5 ± 13.4^{b}	106.3 ± 9.1^{c}	102.9 ± 6.9^{c}	96.2 ± 9.8^{e}	90.4 ± 7.8^{ae}	93.9 ± 7.8^{e}	87.2 ± 6.5^{a}	86.9 ± 8.5^{a}
BMI (kg·m^{-2})	36.8 ± 2.8^{d}	34.9 ± 4.4^{b}	29.4 ± 2.2^{ce}	30.0 ± 1.8^{c}	29.9 ± 2.7^{c}	25.6 ± 1.9^{a}	26.7 ± 1.9^{a}	26.6 ± 1.8^{a}	25.8 ± 1.7^{a}

*BMI = body mass index (mass in kg·height^{-1} in m^{2}).

†If positions share a letter within each row, they are not significantly different at $p < 0.05$.

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Table 4

Mean (±*SD*) total body composition measures of collegiate football players by position.*†

Position	OL (n = 125)	DL (n = 67)	TE (n = 38)	LB (n = 62)	RB (n = 48)	PK (n = 13)	QB (n = 25)	DB (n = 85)	WR (n = 90)
Total percent fat (%)	29.8 ± 3.4[d]	22.9 ± 6.9[b]	18.5 ± 3.6[e]	17.9 ± 4.2[ce]	15.5 ± 4.3[c]	16.7 ± 3.4[ae]	16.6 ± 4.2[cef]	13.1 ± 2.8[a]	13.9 ± 3.5[af]
Total fat mass (kg)	40.3 ± 6.1[d]	28.7 ± 10.6[b]	19.8 ± 4.9[e]	18.5 ± 4.8[ce]	15.1 ± 5.4[cf]	15.2.3 ± 3.9[ace]	15.6 ± 4.3[cef]	11.5 ± 2.8[a]	12.3 ± 3.7[af]
Total lean mass (kg)	90.0 ± 6.4[b]	89.4 ± 7.5[b]	82.0 ± 6.4[e]	79.9 ± 5.5[ce]	77.5 ± 6.9[cd]	71.3 ± 4.8[a]	73.9 ± 6.3[ad]	72.0 ± 5.4[a]	71.0 ± 6.3[a]
Total BMC (kg)	4.71 ± 0.35[b]	4.70 ± 0.36[b]	4.42 ± 0.37[c]	4.40 ± 0.37[c]	4.13 ± 0.41[a]	3.85 ± 0.28[ad]	4.10 ± 0.42[ad]	4.00 ± 0.40[ad]	3.95 ± 0.41[ad]
Total BMD (g·cm^{-2})	1.65 ± 0.10[bc]	1.68 ± 0.12[b]	1.60 ± 0.10[cd]	1.64 ± 0.11[bc]	1.60 ± 0.11[cd]	1.48 ± 0.09[a]	1.54 ± 0.09[ad]	1.56 ± 0.11[ad]	1.53 ± 0.11[a]

*BMC = bone mineral content; BMD = bone mineral density.

†If positions share a letter within each row, they are not significantly different at $p < 0.05$.

age differences between CFP and NFL players. The NFL players were, on average, 24.2 years old, whereas the CFP were, on average, 20.1 years old. Men generally reach their peak BMC and BMD between 23.3 and 26.9 years of age (19). Therefore, NFL players' BMC and BMD may be more developed because of their age and years spent in football-specific training and/or strength training.

Dual x-ray absorptiometry allows for not only total but also regional (i.e., arms, legs, and trunk) assessment of body composition. This study observed a significantly lower trunk MBR in CFP compared with controls. These results are similar to our previous research in NFL players that observed a lower trunk MBR than their age-matched controls. Having a lower trunk MBR may be because of CFP and NFL players having a disproportionately higher trunk BMC relative to their trunk LM, despite both body composition measures being higher than those of the control groups. The higher trunk BMC observed in football players compared with controls may be because of the associated high-impact and odd-impact collisions, which occur regularly in this sport (25).

In contrast with trunk MBR, this study identified no significant differences in leg MBR between CFP and controls. These results are consistent with previous research of NFL players, whose leg MBR was also not significantly different from controls (10). The reason for this lack of difference in leg MBR among CFP and NFL players compared with their respective control groups may be because of the fact that both CFP and NFL players' legs support more total mass. Couple this increased leg mass with their high-impact training and the result would be an increase in the load of the lower body for both the muscle and the bone. Thus, keeping the ratio between muscle and bone consistent.

In this study, we observed no differences in arm MBR between CFP and controls. Although arm LM and BMC were higher in CFP compared with controls, the proportion remained similar to that of the controls, resulting in no difference in arm MBR. By contrast, previous research has demonstrated that NFL players had greater arm MBR than controls (10). Because arms do not support body weight as legs do, bone mass in the arm region may not adapt to the same degree as the legs, resulting in more muscle mass per bone mass in NFL players.

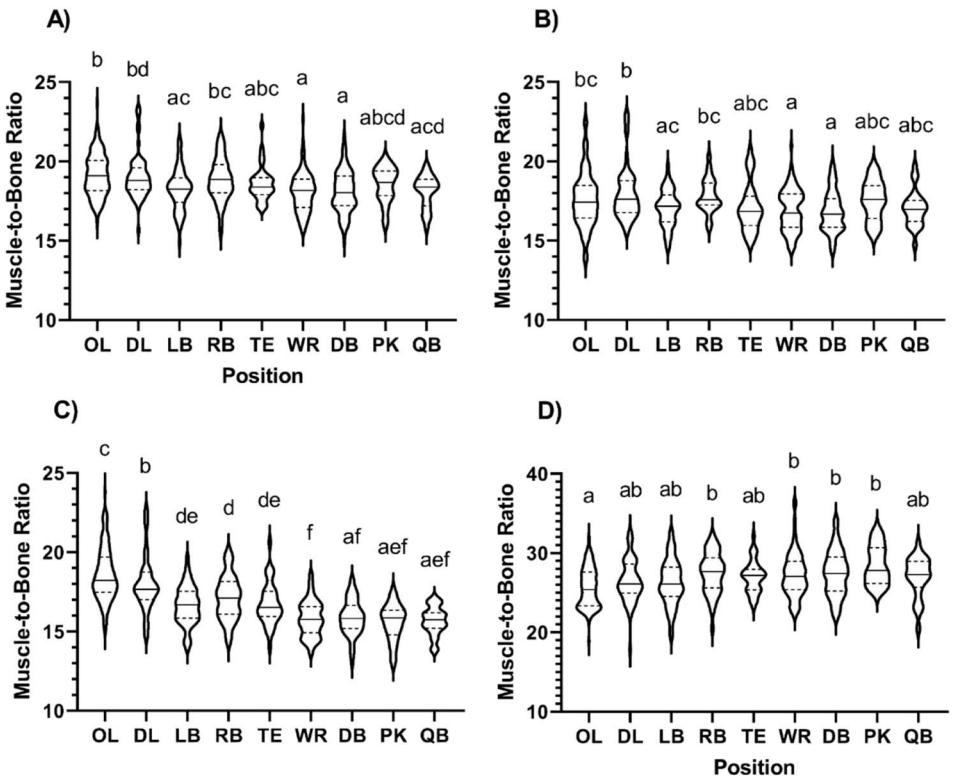

Figure 2. Muscle-to-bone ratio for total (panel A), arms (panel B), legs (panel C), and trunk (panel D) of collegiate football players by position. If positions share a letter within each row, they are not significantly different at $p < 0.05$.

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Table 5

Mean (±*SD*) regional body composition measures of collegiate football players by position.*†

Position	OL (n = 125)	DL (n = 67)	TE (n = 38)	LB (n = 62)	RB (n = 48)	PK (n = 13)	QB (n = 25)	DB (n = 85)	WR (n = 90)
Arm fat mass (kg)	4.1 ± 0.7^{c}	2.9 ± 1.1^{b}	2.1 ± 0.5^{e}	1.9 ± 0.5^{de}	1.6 ± 0.5^{a}	1.5 ± 0.4^{ade}	1.6 ± 0.5^{ade}	1.3 ± 0.3^{a}	1.3 ± 0.4^{a}
Arm lean mass (kg)	12.6 ± 1.4^{b}	13.0 ± 1.6^{b}	11.6 ± 1.4^{c}	11.6 ± 1.2^{c}	11.1 ± 1.1^{cd}	9.8 ± 1.2^{a}	10.4 ± 1.0^{ade}	10.3 ± 1.0^{a}	10.0 ± 1.2^{a}
Arm BMC (kg)	0.72 ± 0.09^{b}	0.73 ± 0.07^{b}	0.68 ± 0.09^{bc}	0.68 ± 0.08^{c}	0.62 ± 0.07^{a}	0.56 ± 0.07^{ad}	0.61 ± 0.07^{ad}	0.61 ± 0.07^{ad}	0.59 ± 0.08^{ad}
Leg fat mass (kg)	12.6 ± 2.5^{c}	10.6 ± 3.6^{b}	7.3 ± 1.9^{e}	6.9 ± 2.0^{de}	5.7 ± 2.0^{df}	5.8 ± 1.5^{aef}	5.6 ± 1.4^{aef}	4.4 ± 1.2^{a}	4.5 ± 1.4^{a}
Leg lean mass (kg)	33.4 ± 2.9^{b}	33.3 ± 3.3^{b}	29.2 ± 2.8^{c}	28.6 ± 2.4^{c}	27.9 ± 2.8^{c}	24.4 ± 1.5^{a}	25.5 ± 2.9^{a}	25.4 ± 2.5^{a}	24.9 ± 2.7^{a}
Leg BMC (kg)	1.80 ± 0.15^{bc}	1.86 ± 0.15^{b}	1.75 ± 0.17^{cf}	1.72 ± 0.15^{cd}	1.63 ± 0.15^{ad}	1.57 ± 0.10^{ae}	1.63 ± 0.19^{adef}	1.60 ± 0.17^{ae}	1.58 ± 0.17^{ae}
Trunk fat mass (kg)	22.5 ± 4.0^{c}	14.1 ± 6.3^{b}	9.5 ± 2.9^{e}	8.7 ± 2.6^{de}	7.0 ± 3.2^{dfg}	7.0 ± 2.0^{aeg}	7.5 ± 2.7^{def}	5.0 ± 1.5^{a}	5.5 ± 2.0^{a}
Trunk lean mass (kg)	40.1 ± 3.1^{b}	39.1 ± 3.3^{bd}	37.6 ± 3.2^{de}	36.0 ± 2.7^{ce}	34.9 ± 3.4^{c}	33.6 ± 2.3^{ac}	34.4 ± 2.9^{ac}	32.7 ± 2.6^{a}	32.7 ± 2.9^{a}
Trunk BMC (kg)	1.58 ± 0.16^{d}	1.49 ± 0.16^{b}	1.40 ± 0.15^{bc}	1.38 ± 0.15^{c}	1.28 ± 0.15^{a}	1.19 ± 0.12^{a}	1.28 ± 0.16^{ac}	1.20 ± 0.15^{a}	1.21 ± 0.16^{a}

*BMC = bone mineral content.

†If positions share a letter within each row, they are not significantly different at $p < 0.05$.

The second aim of this study was to examine positional differences in total and regional MBR in CFP. The results presented here provide new normative data for positional MBR characteristics of collegiate football players that can be used to develop training and nutritional plans. In addition, total and regional positional MBR values in CFP can be compared with those values we have previously reported in NFL players (10).

In this study, we observed numerous differences in MBR by position. This is notable as previous research on NFL players showed no differences in total MBR between the positions (10). The lack of differences in NFL players may be because of the NFL players having more football-specific training over time and increased resources available to athletes (22).

It should be noted that, in this study, we observed no significant differences in total MBR between offensive and defensive positions that mirror each other (i.e., OL vs. DL, LB vs. TE, and DB vs. WR). When comparing mirrored position groups, DB and WR had the lowest total MBR, followed by LB and TE, and OL and DL with the highest total MBR values. The similarities in total MBR between positions that mirror each other may suggest similarities between their on-field physiological demands and position-specific training. Notably, this study also observed no significant differences in total FM, LM, BMC, or BMD between positions that mirror each other. These results are similar to those of past research, which demonstrated that positions that mirror each other have similar total body compositions (6,9,18).

In addition to the assessment of total MBR, this study also explored regional MBR by position. Notably, significant differences in regional MBR between positions were observed, with OL having significantly lower trunk MBR compared with DB, PK, RB, and WR. This contrasts with leg and arm MBR, wherein OL had significantly higher MBR values than all other positions. This finding is similar with previous research in NFL players, which demonstrated OL having significantly lower trunk MBR than PK, suggesting that OL display less LM for the same amount of BMC in the trunk region (10). In addition, DL had significantly higher arm MBR than DB, LB, and WR. Offensive linemen also had significantly higher leg MBR than all other positions. A higher MBR suggests more LM to BMC in that region, or a smaller skeleton. Previous research found that leg MBR values in NFL players were similar in offensive and defensive positions that mirror each other (10), and this study observed the same in CFP.

Strengths of the current study include the use of DXA, the gold standard for body composition, for the determination of MBR. Former studies (3,7,8,13,17) in this area relied on anthropometrics for the determination of lean muscle and bone masses. The use of DXA eliminated the use of a constant for the determination of bone density to determine the bone mass (2,14). In addition, the use of allowed for the determination of not only total but also regional measures of MBR. This study does have a few limitations: Although precautions were taken to control differences between sites (each site used standard best practices), there is the potential for bias based on scan location. In addition, scanning a team in a feasible amount of time limits the control of time of day for scanning, nutrition, hydration, and the like. Each university involved in this study used best practices, and when possible, similar position groups were scanned at the same time of day. However, the large sample size involved in this study should limit the influence of these biases. Finally, this study was cross-sectional in the assessment of body composition and does not address the change over time in collegiate athletes.

Practical Applications

This study explored the use of DXA to determine total and regional MBR in CFP. Although an MBR range suggesting optimal muscular loading of the skeleton may exist, there is currently no normative data to identify such a range in CFP specifically. Thus, this article provides insight into the MBR across positions in CFP, as well as in comparison to a control population. Moreover, these data provide information regarding total and regional MBR for different positions. These data can be used by athletes and coaches as reference data when using DXA to measure total and regional MBR because these numbers will likely differ from others such as skinfold, girth, and circumference measures. Moreover, these data provide information regarding the total and regional MBR for each position. The similarity between mirrored positions observed in this study and our previous report on NFL athletes suggests that MBR may play a role in on-field positional requirements. The data presented in this report can be used to identify ideal ranges for each position in collegiate football players. However, it is important to clarify that the data presented here are averages and *SD* for each position. In practice, performance metrics must be considered because each individual will have their own optimal body type to maximize performance. In collegiate football players total and regional MBR provides an objective measure to monitor changes in muscle and bone masses. As athletes are encouraged to alter their body composition, the MBR can provide a matrix to monitor these alterations. This article provides normative data to compare players against the normal population and positional differences that may prove to be useful in evaluating players.

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